ML FLOW WITH CI/CD Pipeline

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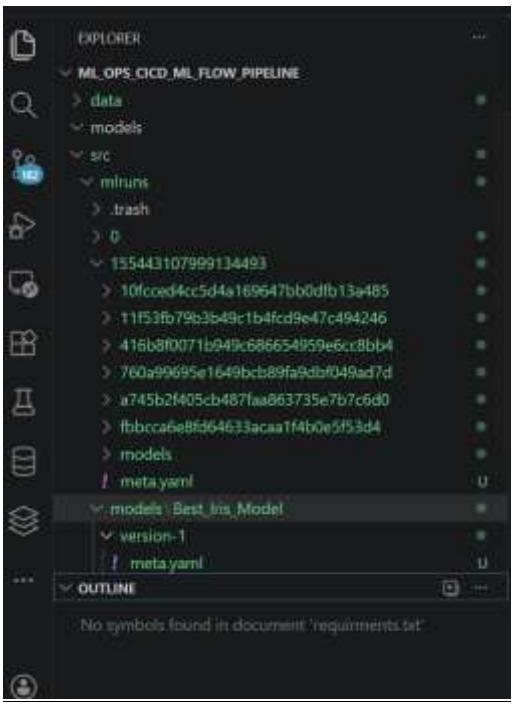
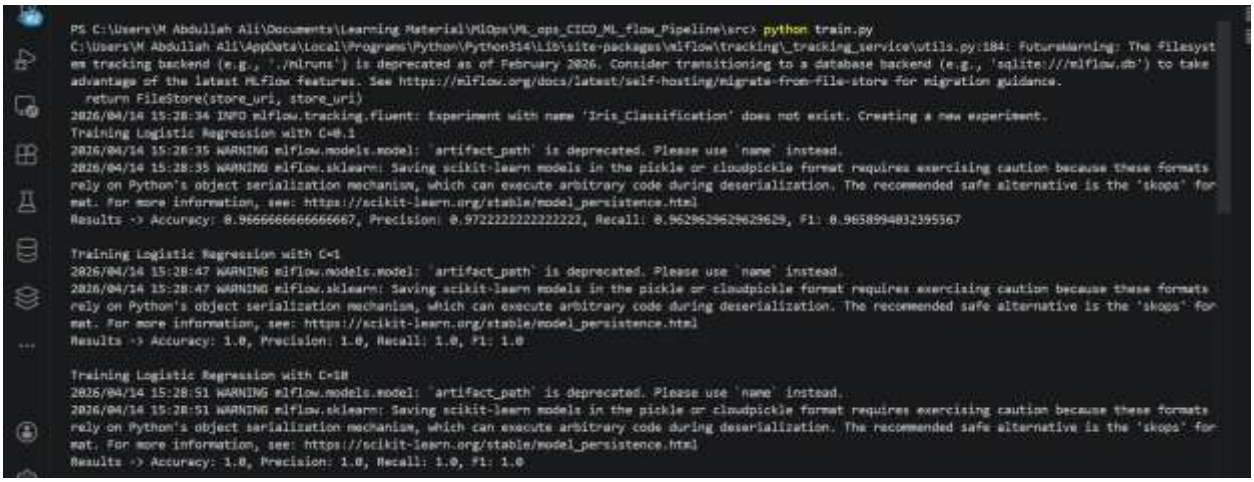
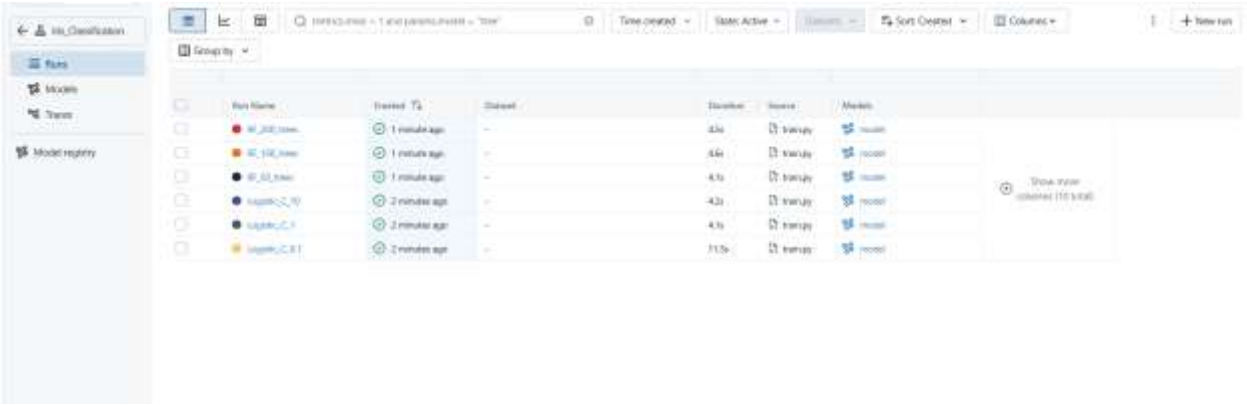
PART 1

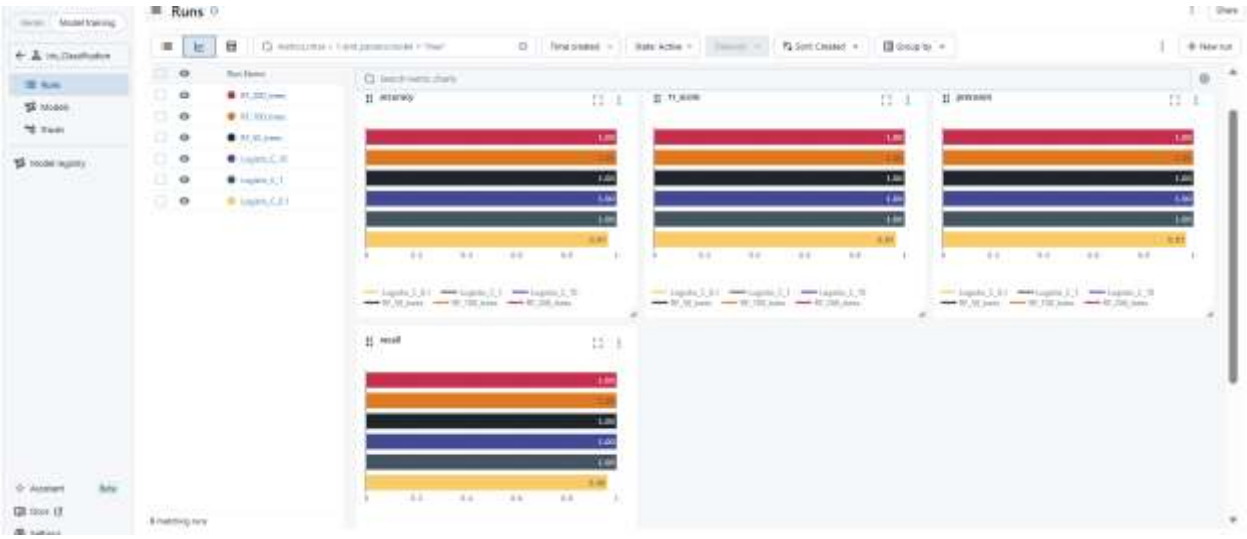
Model Traing and ML flow runs

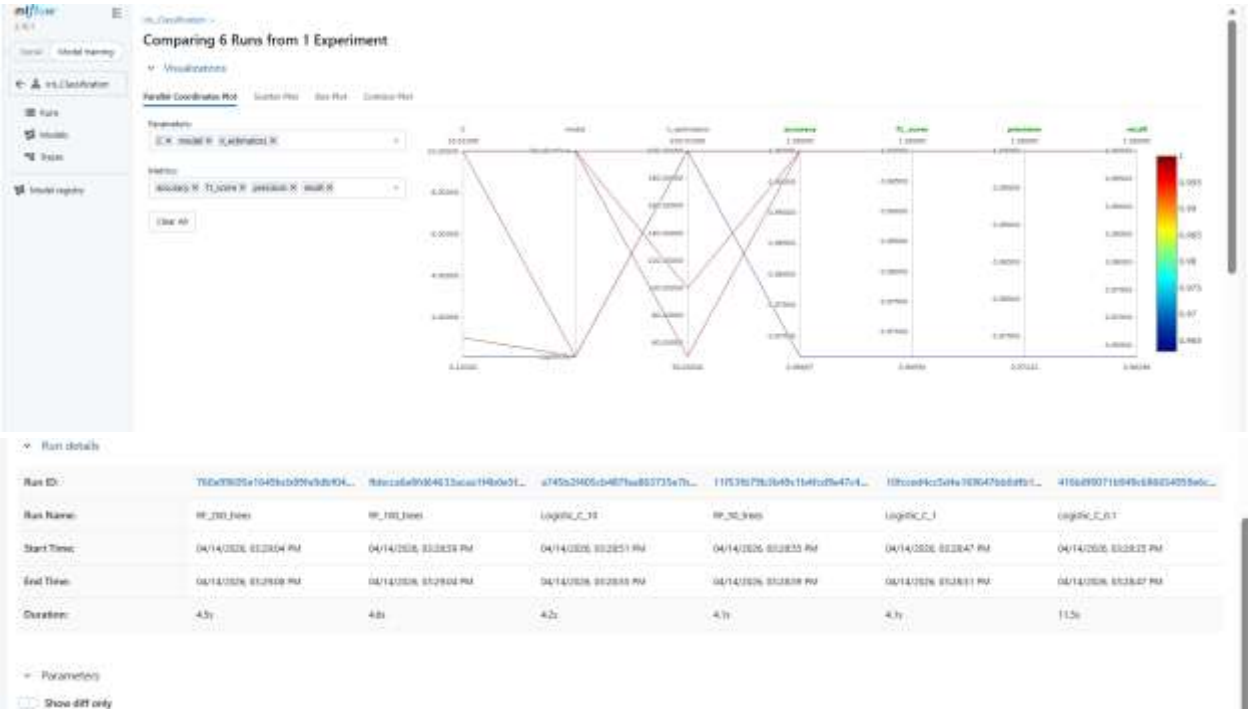


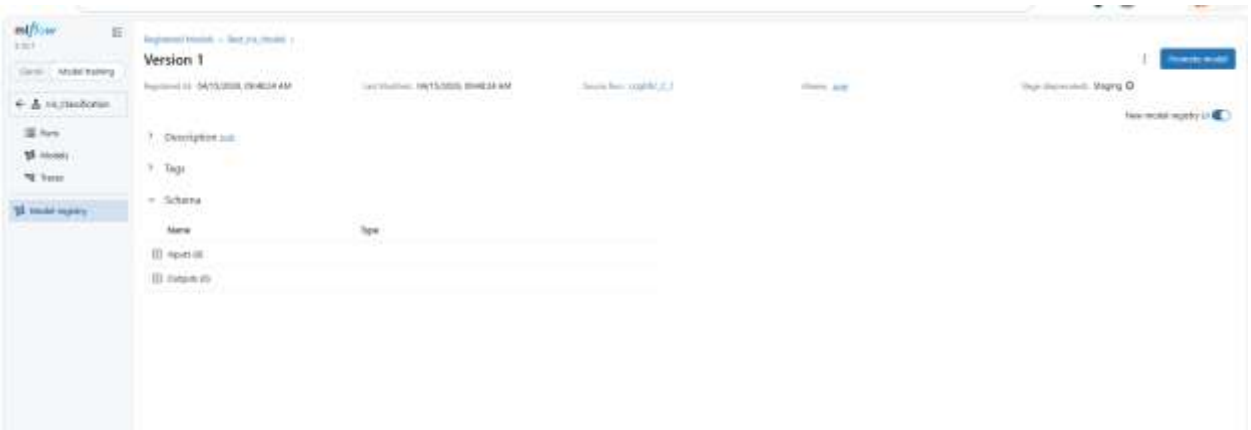


```
Training Random Forest with n_estimators=50
2026/04/14 15:28:55 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
2026/04/14 15:28:55 WARNING mlflow.sklearn: Saving scikit-learn models in the pickle or cloudpickle format requires exercising caution because these formats
rely on Python's object serialization mechanism, which can execute arbitrary code during deserialization. The recommended safe alternative is the 'skops' for
mat. For more information, see: https://scikit-learn.org/stable/model_persistence.html
Results -> Accuracy: 1.0, Precision: 1.0, Recall: 1.0, F1: 1.0

Training Random Forest with n_estimators=100
2026/04/14 15:28:59 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
2026/04/14 15:28:59 WARNING mlflow.sklearn: Saving scikit-learn models in the pickle or cloudpickle format requires exercising caution because these formats
rely on Python's object serialization mechanism, which can execute arbitrary code during deserialization. The recommended safe alternative is the 'skops' for
mat. For more information, see: https://scikit-learn.org/stable/model_persistence.html
Results -> Accuracy: 1.0, Precision: 1.0, Recall: 1.0, F1: 1.0

Training Random Forest with n_estimators=200
Model: LogisticRegression
Parameters: {'C': 1}
Best F1 Score: 1.0
C:\Users\M Abdullah Ali\AppData\Local\Programs\Python\Python314\lib\site-packages\mlflow\tracking\_model_registry\utils.py:220: FutureWarning: The filesystem
 model registry backend (e.g., './mlruns') is deprecated as of February 2026. Consider transitioning to a database backend (e.g., 'sqlite:///mlflow.db') to t
ake advantage of the latest MLflow features. See https://mlflow.org/docs/latest/self-hosting/migrate-from-file-store for migration guidance.
  return FileStore(store_uri)
Successfully registered model 'Best_Iris_Model'.
2026/04/14 15:29:08 WARNING mlflow.tracking._model_registry.fluent: Run with id 10fcced4cc5d4a169647bb0df613e485 has no artifacts at artifact path 'model', r
egistering model based on models:/m-98d14441ae7d4b53a479b52820919ac0 instead
Created version '1' of model 'Best_Iris_Model'.
C:\Users\M Abdullah Ali\Documents\Learning Material\MLOps\ML_ops_CICD_ML_flow_Pipeline\src\train.py:125: FutureWarning: ``mlflow.tracking.client.MlflowClient
```









<u>Explanation:</u>

Here is a **short, assignment-style explanation based on your case (Logistic Regression C=1 as best model):**

<u>Performance Analysis</u>

<u>Accuracy</u>

Logistic Regression with C=1 achieved an accuracy of 1.00, meaning all test samples were correctly classified. Other models also performed similarly, but C=0.1 showed slightly lower accuracy (~0.97) due to stronger regularization

Precision

For C=1, precision is 1.00, indicating that all predicted classes are correct. There are no false positive predictions in this case

Recall

Recall for C=1 is also 1.00, which means the model successfully identified all actual class instances without missing any

F1-Score

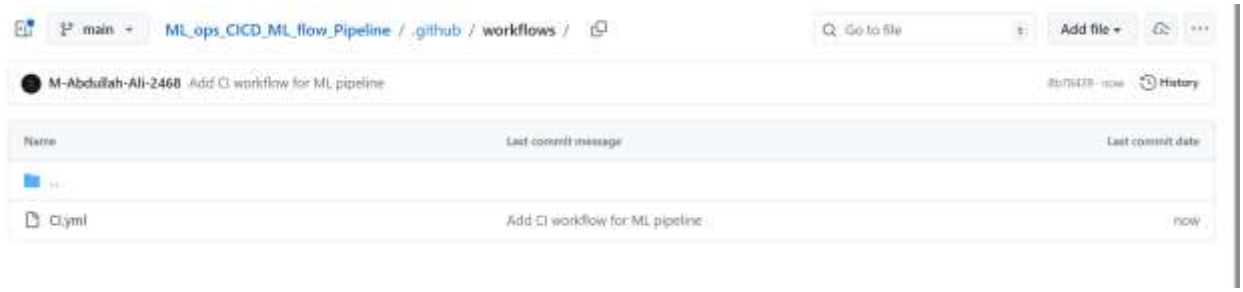
The F1-score for C=1 is 1.00, confirming a perfect balance between precision and recall. This shows the model is both accurate and consistent.

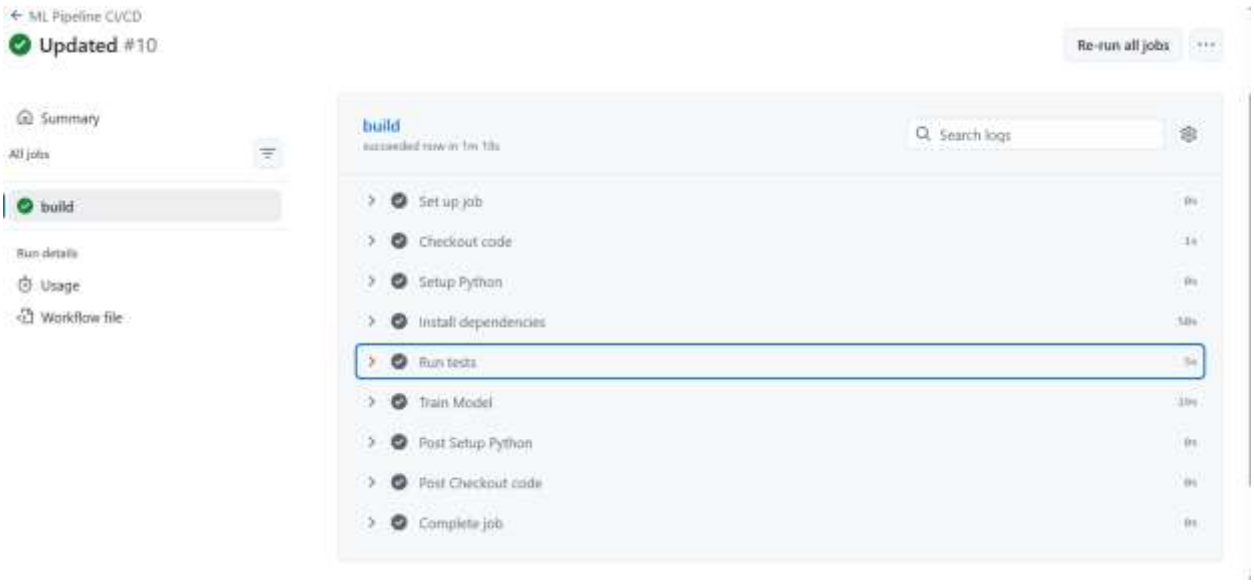
conclusion

Logistic Regression with C=1 is selected as the best model because it achieves perfect performance across all evaluation metrics while maintaining a simple and efficient structure. Compared to C=0.1, it avoids underfitting, and it performs equally well as more complex models like Random Forest.

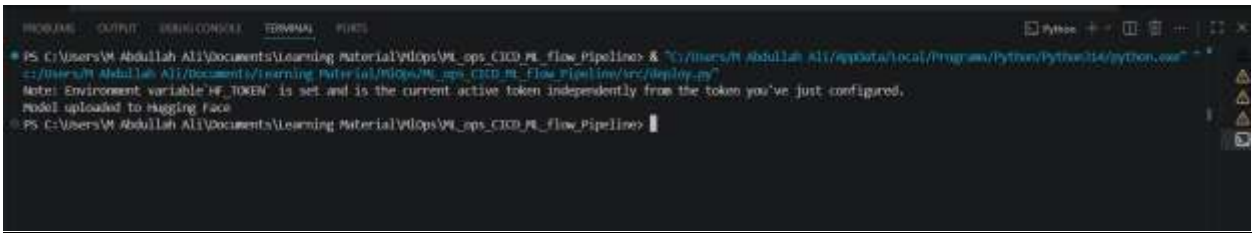
Part 2: GitHub CI/CD Workflow

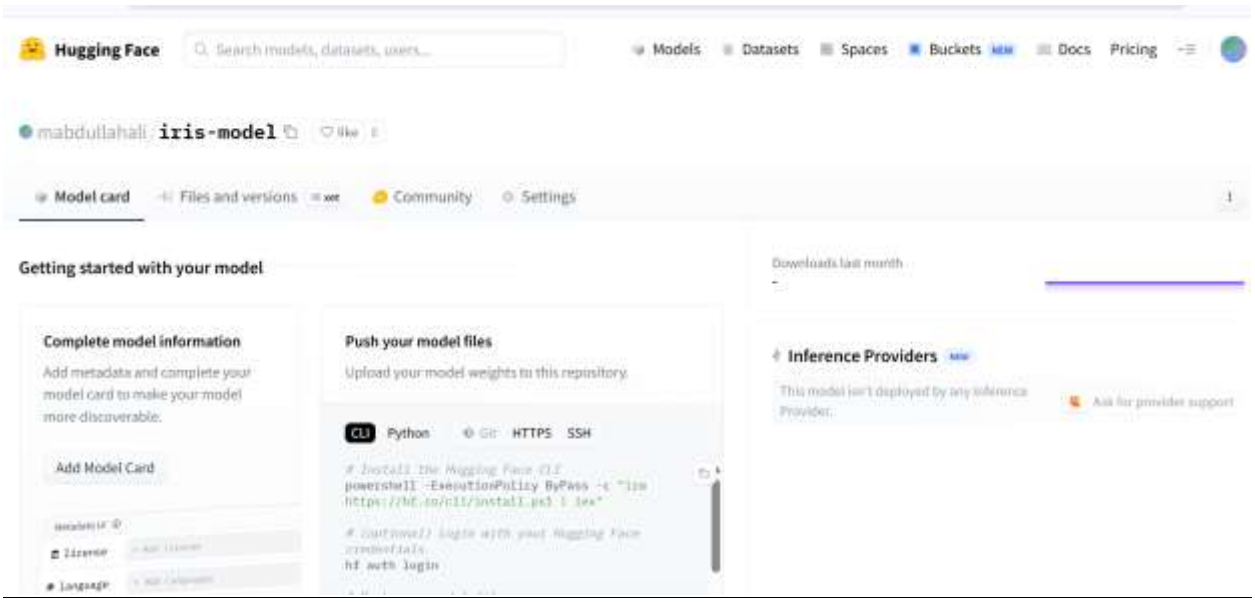
Workflow Created:




ML Pipeline CI/CD
Updated #10
Re-run all jobs
Summary
All jobs
build
Run details
Usage
Workflow file
build
Search logs
Set up job
Checkout code
Setup Python
Install dependencies
Run tests
Train Model
Post Setup Python
Post Checkout code
Complete job

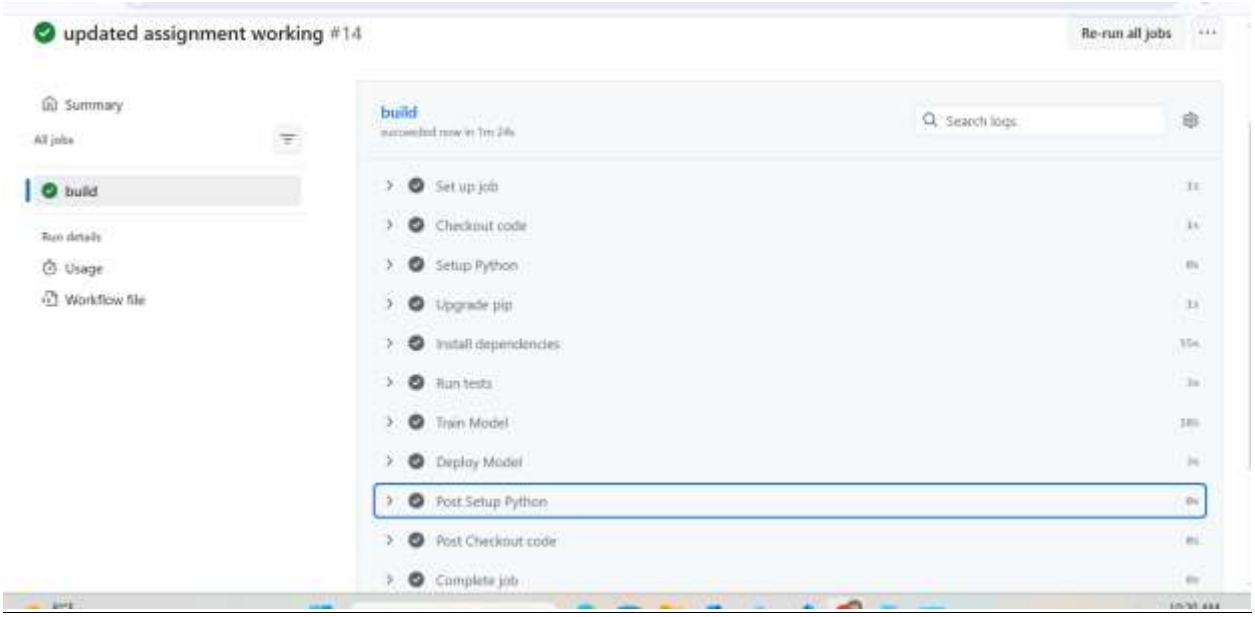
Deployment:


PS C:\Users\M Abdullah Ali\Documents\Learning Material\MlOps\ML_ops_CICD_ML_flow_Pipeline> & "C:/Users/M Abdullah Ali/AppData/Local/Programs/Python/Python314/python.exe" -*
c:/Users/M Abdullah Ali/Documents/Learning Material/MlOps/ML_ops_CICD_ML_flow_Pipeline/src/deploy.py"
Note: Environment variable`HF_TOKEN` is set and is the current active token independently from the token you've just configured.
Model uploaded to Hugging Face
PS C:\Users\M Abdullah Ali\Documents\Learning Material\MlOps\ML_ops_CICD_ML_flow_Pipeline>


Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets
Docs
Pricing
mabdullahali iris-model
Model card
Files and versions
Community
Settings
Getting started with your model
Downloads last month
Complete model information
Add metadata and complete your model card to make your model more discoverable.
Add Model Card
Push your model files
Upload your model weights to this repository.
CLI
Python
Git
HTTPS
SSH
Install the Hugging Face CLI
powershell -ExecutionPolicy ByPass -c "irm https://hf.co/cli/install.ps1 | iex"
(optional) Login with your Hugging Face credentials
hf auth login
Inference Providers
This model isn't deployed by any Inference Provider.
Ask for provider support
License
Language

Model staging changed to production:

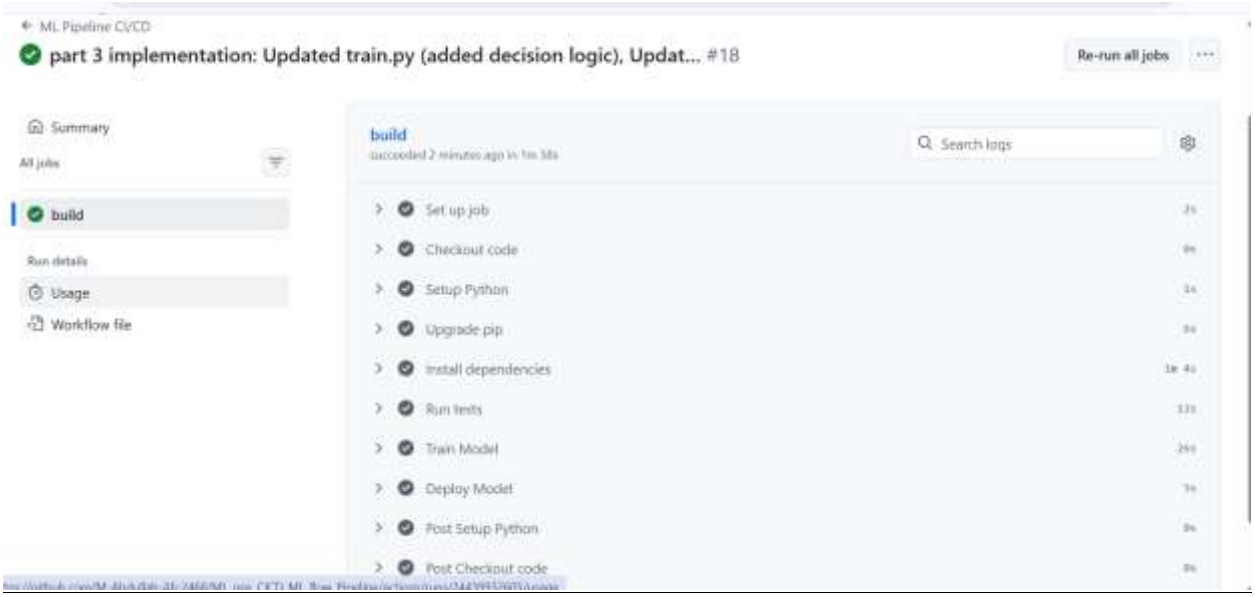
Workflow updated:



Part 3: Auto Retraining Modulefi

Model retraining runs:

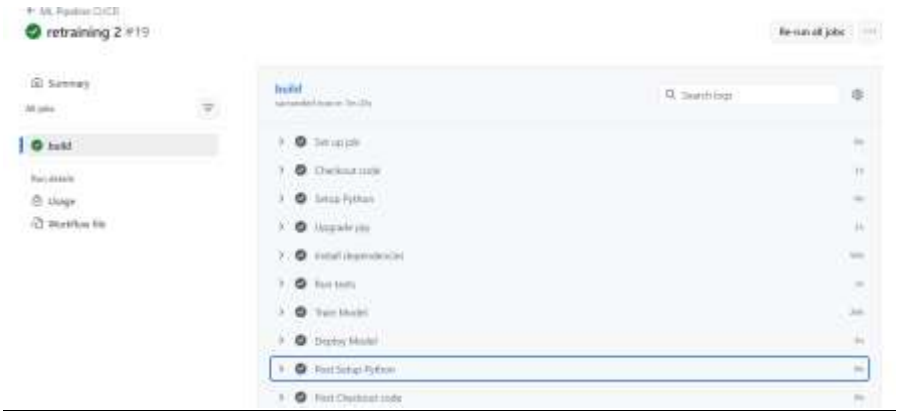
Run1:




```
our migration guide here: https://mlflow.org/docs/latest
  client.transition_model_version_stage(
Logistic C=0.1 -> F1: 0.9658994032395567
Logistic C=1 -> F1: 1.0
Logistic C=10 -> F1: 0.9658994032395567
RF n=50 -> F1: 1.0
RF n=100 -> F1: 1.0
RF n=200 -> F1: 1.0
Best model saved at: models/best_model.pkl
No registered model found -> first run
Old Production F1: 0
New Model F1: 1.0
New model is better -> promoting
Model version 1 moved to Staging
```

```
New Data Upload              : 100%|██████████|   975B /   975B, 1.63kB/s

  best_model.pkl             : 100%|██████████|   975B /   975B
Model uploaded to Hugging Face
```

Run2: Retraining again




```
our migration guide here: https://mlflow.org/docs/latest/model-registry.html#mig
  client.transition_model_version_stage(
Logistic C=0.1 -> F1: 0.9658994032395567
Logistic C=1 -> F1: 1.0
Logistic C=10 -> F1: 0.9658994032395567
RF n=50 -> F1: 1.0
RF n=100 -> F1: 1.0
RF n=200 -> F1: 1.0
Best model saved at: models/best_model.pkl
No registered model found -> first run
Old Production F1: 0
New Model F1: 1.0
New model is better -> promoting
Model version 1 moved to Staging
```

```
  client.transition_model_version_stage(
Using Staging model version: 1
Model promoted to Production
Processing Files (0 / 0)      : |          |  0.00B /  0.00B
New Data Upload               : |          |  0.00B /  0.00B
  best_model.pkl              : 100%|########|   975B /   975B
  best_model.pkl              : 100%|########|   975B /   975B
Processing Files (1 / 1)      : 100%|########|   975B /   975B,   ???B/s
New Data Upload               : 100%|########|   975B /   975B,   ???B/s
  best_model.pkl              : 100%|########|   975B /   975B
Processing Files (1 / 1)      : 100%|########|   975B /   975B,  0.00B/s
New Data Upload               : 100%|########|   975B /   975B,  0.00B/s
  best_model.pkl              : 100%|########|   975B /   975B
Model uploaded to Hugging Face
```

<u>Explanation of Assignment:</u>

Great 👍 — ab main tumhe **har part ki pipeline (short + clear)** bhi add karke de raha hoon, assignment style me.

Part 1: MLflow Complete Workflow

Work Done:

- Selected Iris dataset for classification.
- Trained Logistic Regression and Random Forest with different parameters.
- Applied preprocessing (train-test split and scaling).
- Logged parameters, metrics, and models using MLflow.
- Compared multiple runs in MLflow UI.
- Registered the best model and assigned Staging stage.

Pipeline:

Dataset → Preprocessing → Train Multiple Models → Evaluate → Log to MLflow → Compare → Register Best Model

Part 2: GitHub CI/CD Workflow

Work Done:

- Organized project structure (src, models, requirements).
- Created GitHub Actions workflow to run on push.
- Installed dependencies and executed training automatically.

- Added pytest for testing.
- Automated training, logging, and artifact saving.
- Deployed best model to Hugging Face.

Pipeline:

Code Push → GitHub Actions → Install Dependencies → Run Tests → Train Model → Log to MLflow → Deploy Model

Part 3: Auto Retraining Module

Work Done:

- Implemented scheduled retraining using cron (weekly).
- Simulated new incoming data for retraining.
- Compared new model with production model using F1-score.
- Promoted model only if performance improved.
- Handled first-run case when no model exists.
- Ensured only one active model in staging/production.
- Used SQLite database for persistent MLflow tracking.
- Demonstrated multiple retraining cycles.

Pipeline:

Scheduled Trigger / New Data → Retrain Model → Log to MLflow → Compare with Production →

IF Better → Promote (Staging → Production) → Deploy

ELSE → Skip